

Publications

I have published two monographs, one textbook and three invited chapters, more than 100 papers up to now, including 51 SCI journal papers. The publication list is shown as follows:

Monographs and Book Chapters

1. **Wang, X.**, Wang, X., Zhai, W., and Cai, K. (2024). Sparse Sensing and Sparsity Sensed in Multi-Sensor Array Applications, Publisher: Springer Nature, 2023.
2. **Wang, X.** and X. Cao, Advanced Digital Signal Processing: From Theory to Applications, Publisher: Press of Beihang University, 2022.
3. Cui, G., Liang, J., Liao, B., **Wang, X.**, and Kong, L. (Eds.). (2024). Radar Array Design Using Optimization Theory. SciTech Publishing Inc.
4. **X. Wang**, H. Liu, M. Amin, K. Cai and Y. Zhu, Y. A Hand Air-Writing System Using MIMO Radar and Deep Learning. In Radar-Based Human-Computer Interaction (pp. 153-194). Cham: Springer Nature Switzerland.
5. **X. Wang**, Detection and Tracking of UAVs using Interferometric Radar” in Book, Radar Countermeasures for Unmanned Aerial Vehicles, Publisher: IET Press, 2021.
6. **X. Wang**, Sparse MIMO Array Transceiver Design in Dynamic Environment in Book, Sparse Arrays for Radar, Sonar, and Communication, Publisher: IET Press/Wiley, 2023.

Journals Published

1. **Xiangrong Wang**, Elias Aboutanios, Mathew Trinkle, Moeness Amin. “Reconfigurable Adaptive Array Beamforming by Antenna Selection.” *IEEE Transactions on Signal Processing*, 2014, Vol. 62, No. 9, pp. 2385-2396.
2. **Xiangrong Wang**, Elias Aboutanios, Moeness Amin. “Thinned Array Beam pattern Synthesis by Iterative Soft-Thresholding based Optimization Algorithms.” *IEEE Transactions on Antennas and Propagation*, 2014, Vol. 62, No. 12, pp. 6102-6113.
3. **Xiangrong Wang**, Elias Aboutanios, Moeness Amin. “Adaptive Array Thinning for Enhanced DOA Estimation.” *IEEE Signal Processing Letters*, 2015, Vol. 22, No. 7,

pp. 799-803.

4. **Xiangrong Wang**, Elias Aboutanios, Moeness Amin. "Reduced-Rank STAP for Slow Moving Target Detection by Antenna-Pulse Selection." *IEEE Signal Processing Letters*, 2015, Vol. 22, No. 8, pp. 1156-1160.
5. Moeness Amin, **Xiangrong Wang**, Yimin Zhang, Fauzia Ahmad, Elias Aboutanios. "Sparse Arrays and Sampling for Interference Mitigation and DOA Estimation in GNSS." *The Proceedings of IEEE*, 2016, Vol. 104, No. 6, pp. 1302-1317.
6. **Xiangrong Wang**, Elias Aboutanios. "Slow Radar Target Detection in Heterogeneous Clutter using Thinned STAP." *IET Radar, Sonar and Navigation*, 2016, Vol. 10, No. 4, pp. 726-734.
7. **Xiangrong Wang**, Moeness Amin, Xianghua Wang and Xianbin Cao. "Sparse Array Quiescent Beamformer Design Combining Adaptive and Deterministic Constraints." *IEEE Transactions on Antennas and Propagation*, 2017, Vol. 65, No. 11, pp. 5808-5818.
8. **Xiangrong Wang**, Moeness Amin, Fauzia Ahmad, Elias Aboutanios. "Interference DOA Estimation and Suppression for GNSS Receivers using Fully Augmentable Arrays." *IET Radar, Sonar and Navigation*, 2017, Vol. 11, No. 3, pp. 474-480.
9. **Xiangrong Wang**, Moeness Amin and Xianbin Cao. "Analysis and Design of Optimum Sparse Array Configurations for Adaptive Beamforming." *IEEE Transactions on Signal Processing*, 2018, Vol. 66, No. 2, pp. 340-351.
10. **Xiangrong Wang**, Moeness Amin and Xianghua Wang. "Robust Sparse Array Design for Adaptive Beamforming Against DOA Mismatch." *Signal Processing*, 2018, Vol. 146, No. 5, pp. 41-49.
11. **Xiangrong Wang**, Aboulnasr Hassanien and Moeness Amin. "Sparse Transmit Array Design for Dual-Function Radar Communications by Antenna Selection." *Digital Signal Processing*, 2018, Vol. 83, pp. 223-234.
12. **Xiangrong Wang**, Pengcheng Wang, Xianbin Cao and Victor C. Chen. "Interferometric angular velocity measurement of rotating blades: theoretical analysis, modelling and simulation study." *IET Radar, Sonar and Navigation*, 2019, Vol. 13, No. 3, pp. 438-444.

13. **Xiangrong Wang**, Aboulnasr Hassanien and Moeness Amin. "Dual-Function MIMO Radar Communications System Design Via Sparse Array Optimization." *IEEE Transactions on Aerospace and Electronic Systems*, 2019, Vol. 55, No. 3, pp. 1213-1226.
14. **Xiangrong Wang**, Pengcheng Wang and Victor C. Chen. "Simultaneous Measurement of Radial and Transversal Velocities using Interferometric Radar." *IEEE Transactions on Aerospace and Electronic Systems*, 2020, Vol. 56, No. 4, pp. 3080-3098.
15. **Xiangrong Wang** and Elias Aboutanios. "Sparse array design for multiple switched beams using iterative antenna selection method." *Digital Signal Processing*, 2020, Vol. 105, pp. 102684.
16. **Xiangrong Wang**, Fauzia Ahmad, Kutlu Dogancay and Sundeepr Prabhakar Chepuri. "Editorial for special issue on Optimum Sparse Arrays and Sensor Placement for Environmental Sensing." *Digital Signal Processing*, 2020, Vol. 105, pp. 102839.
17. **Xiangrong Wang**, Weitong Zhai and Alfonso Farina. "A Unified Framework of Adaptive Sidelobe Canceller Design by Antenna/Subarray Selection." *Signal Processing*, Vol. 189, pp. 109283, Dec 2021.
18. Yanan Ma, Xianbin Cao, **Xiangrong Wang**. "Multi-source Off-Grid DOA Estimation using Iterative Phase Offset Correction in Coarray Domain." *Digital Signal Processing*, Vol. 112, May 2021.
19. Xuan Zhang **Xiangrong Wang**, Hing Cheung So. "Linear Array Pattern Synthesis with Shape Constraints and Excitation Range Control." *IEEE Antennas and Wireless Propagation Letters*, 2021, Vol. 20, No. 6, pp. 1018-1022.
20. Xuan Zhang and **Xiangrong Wang**. "Waveform Design with Controllable Modulus Dynamic Range Under Spectral Constraints." *Signal Processing*, vol. 189, pp. 108285, Dec 2021.
21. Yanan Ma, Xianbin Cao, **Xiangrong Wang**, Maria S. Greco, and Fulvio Gini. "Multi-source Off-grid DOA Estimation with Single Snapshot using Arbitrary Linear Arrays." *Signal Processing*, Vol. 189, pp. 108238, Signal Processing, 2021.
22. Saima Ishtiaq, **Xiangrong Wang** and Shahid Hassan. "Multi-Target Tracking Algorithm Based on 2-D Velocity Measurements Using Dual-Frequency Interferometric Radar." *Electronics*, Vol. 10, No. 16, Article No. 1969, 2021.
23. **Xiangrong Wang**, Maria Greco, Fulvio Gini. "Adaptive Sparse Array Beamformer Design by Regularized Complementary Antenna Switching." *IEEE Transactions on*

Signal Processing, 2021, Vol. 69, pp. 2302-2315.

24. Weitong Zhai, **Xiangrong Wang**, Maria Greco and Fulvio Gini, "Joint Optimization of Sparse FDAs for Time Invariant Transmit Beampattern Synthesis", *IEEE Signal Processing Letter*, Vol. 29, pp. 110-114, 2022.
25. Qinghui Lu, Guolong Cui, Ruitao Liu, Xianxiang Yu, **Xiangrong Wang** and Lingjiang Kong, "Wideband Beampattern Synthesis using Single Digital Beamformer with Integer Time Delay Filters", *IEEE Transactions on Antennas and Propagation*, vol. 70, no. 5, pp. 3437-3449, May 2022.
26. Hamed Nosrati, Elias Aboutanios, **Xiangrong Wang** and David Smith, "Switch-based Hybrid Beamforming for Massive MIMO Communications in mmWave Bands", *Signal Processing*, vol. 200, Nov, 2022, doi:10.1016/j.sigpro.2022.108659.
27. **Xiangrong Wang**, Weiliang Li and Victor C. Chen, "Hand Gesture Recognition Using Radial and Transversal Dual Micro-Motion Features", *IEEE Transactions on Aerospace and Electronic Systems*, vol. 58, no. 6, pp. 5963-5973, Dec. 2022.
28. **Xiangrong Wang**, Weitong Zhai, Maria Greco and Fulvio Gini, "Cognitive Sparse Beamformer Design in Dynamic Environment via Regularized Switching Network", *IEEE Transactions on Aerospace and Electronic Systems*, vol. 59, no. 2, pp. 1816-1833, April 2023.
29. Jing Xu, **Xiangrong Wang**, Elias Aboutanios and Guolong Cui, "Hybrid Index Modulation for Dual-functional Radar Communications Systems", *IEEE Transactions on Vehicular Technology*, vol. 72, no. 3, pp. 3186-3200, March 2023.
30. Xuan Zhang, **Xiangrong Wang**, and Xianghua Wang, "Joint Antenna Selection and Waveform Design for Coexistence of MIMO Radar and Communications", *EURASIP Journal on Advances in Signal Processing*, 127(2022), 2022.
31. Saima Ishtiaq, **Xiangrong Wang**, Shahid Hassan, Alsharef Mohammad, Ahmad Aziz Alahmadi and Nasim Ullah "Three-Dimensional Multi-Target Tracking using Dual-Orthogonal Baseline Interferometric Radar", *Sensors*, 22(19):7549, 2022.
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33. Weitong Zhai, **Xiangrong Wang**, Maria Greco and Fulvio Gini, "Reinforcement Learning based Dual-Functional Massive MIMO Systems for Multi-target Detection

- and Communications”, *IEEE Transactions on Signal Processing*, vol. 71, pp. 741-755, 2023.
34. Hassan Shahid, **Xiangrong Wang**, Saima Ishtiaq, Nasim Ullah, Alsharef Mohammad, and Abdulfattah Noorwali. “Human Activity Classification Based on Dual Micro-Motion Signatures Using Interferometric Radar”, *Remote Sensing* 15(7) 1752, 2023.
 35. **Xiangrong Wang**, Weitong Zhai, Xuan Zhang, Xianghua Wang and Moeness Amin, “Enhanced Automotive Sensing Assisted by Joint Communication and Cognitive Sparse MIMO Radar”, *IEEE Transactions on Aerospace and Electronic Systems*, vol. 59, no. 5, pp. 4782-4799, Oct. 2023.
 36. Xuan Zhang, **Xiangrong Wang**, Jiayi Huang and Hing Cheung So, “Joint Design of Antenna Selection and Transmit Beamformer for Radar-Communications Integration”, *Digital Signal Processing*, Volume 144, January 2024.
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 38. Jing Xu, **Xiangrong Wang**, Xuan Zhang, Xianghua Wang and Abdelhak M. Zoubir, “Joint Design of Transmit Precoding and Antenna Selection for Multiple-User Multiple-Target MIMO DFRC”, *Signal Processing*, Vol. 218, pp. 109364, May 2024.
 39. Xianghua Wang, Youqing Wang, Ziyi Zhang, **Xiangrong Wang** and Ron Patton, “Sensor Fault Tolerant Control for a 3-DOF Helicopter Considering Detectability Loss,” in *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 70, no. 10, pp. 4112-4125, Oct. 2023.
 40. W. Wei, G. Cui, X. Yu, R. Liu and **X. Wang**, “Asymmetric Beampattern Synthesis for Rectangular Planar Array via Window Function Design,” in *IEEE Transactions on Signal Processing*, vol. 72, pp. 400-414, 2024.
 41. Xuan Zhang, **Xiangrong Wang**, and Moeness Amin, “Transmit Waveform Design for Integrated Wideband MIMO Radar and Bi-directional Communications”, *IEEE Transactions on Vehicular Technology*, vol. 73, no. 9, pp. 13482-13497, Sept. 2024, doi: 10.1109/TVT.2024.3386755.
 42. **Xiangrong Wang**, Weitong Zhai, Xianghua Wang, Moeness G. Amin and Kaiquan Cai, “Wideband Near-Field Integrated Sensing and Communication With Sparse Transceiver

Design,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 18, no. 4, pp. 662-677, May 2024, doi: 10.1109/JSTSP.2024.3394970.

43. **Xiangrong Wang**, Hengfeng Liu, Xianghua Wang, Victor C. Chen, Moeness G. Amin and Kaiquan Cai, “Sensing, Tracking, and Recognition of Macro–Micro Hand Gestures Using Interferometric MIMO Radar,” *IEEE Transactions on Instrumentation and Measurement*, vol. 73, pp. 1-14, 2024, Art no. 8004314, doi: 10.1109/TIM.2024.3412196.
44. **Xiangrong Wang**, Kun Wu, Xianghua Wang, Moeness Amin and Abdelhak Zoubir, “Synthetic Interferometry Exploiting Radar Motion”,*IEEE Geoscience and Remote Sensing Letters*, vol. 21, pp. 1-5, 2024, Art no. 3510605, doi: 10.1109/LGRS.2024.3477711.
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48. W. Zhai,**X. Wang**, C. Zhou, M. S. Greco, F. Gini and Z. Shi, “Dual-Functional IRS Design for ISAC via Reinforcement Learning in Unknown Environment,” in *IEEE Transactions on Aerospace and Electronic Systems*, vol. 61, no. 6, pp. 17785-17803, Dec. 2025.
49. **X. Wang**, F. Gini and K. Cai, “Reconfigurable Integrated Sensing and Communications (RISAC): Sparse MIMO and Hybrid Beamforming in Far and Near Fields,” in *Proceedings of the IEEE*, vol. 114, no. 1, pp. 93-120, Jan. 2026, doi: 10.1109/JPROC.2026.3670886.
50. X. Wang, C. P. Tan, Y. Wang and **X. Wang**, “Active Fault-Tolerant Control of a Prototype Helicopter Against Actuator and Sensor Faults,” in *IEEE Transactions on Industrial Electronics*, vol. 73, no. 3, pp. 4713-4724, March 2026, doi: 10.1109/TIE.2025.3613640.
51. W. Zhai, **X. Wang**, J. Zhang, J. Xie, K. Cai and X. Cao, “Optimal Sparse Conformal Array Design for Dual-Polarized Weather Radar with Integrated Communications in Low-Altitude Airspace,” in *Science China-Information Sciences*, accepted.

Journals In Progress

1. W. Zhai, **X. Wang**, K. Cai, Y. Zhu and Y. Eldar, “Generalized Beamforming for Dual-Functional MIMO Radar and Multi-user Communications”, submitted to *IEEE Transactions on Wireless Communications*, under review.
2. W. Zhai, **X. Wang**, M. Greco, F. Gini and R. Zhang, “RIS-aided ISAC Design with Low-Resolution DAC/ADC and Discrete Phase Shifters”, submitted to *IEEE Transactions on Signal Processing*, under review.
3. H. Liu, **X. Wang**, K. Cai, X. Wang and M. Amin, “Enhanced Omnidirectional Human Activity Recognition via Modified High-Dimensional Model Representations”, submitted to *IEEE Transactions on Radar Systems*, under review.
4. S. Zhang, **X. Wang**, J. Liu and M. Amin, “Geometric Multipath Exploitation for Mixed LOS and NLOS Target Localization using Single-Channel Radar”, submitted to *IEEE Transactions on Aerospace and Electronic Systems*, under review.

Conference Proceedings

1. **Xiangrong Wang**, Elias Aboutanios. “Reconfigurable Adaptive Linear Array Signal Processing in GNSS Applications.” *Proceedings of the 38th IEEE ICASSP*, May 2013, pp. 4154-4158.
2. **Xiangrong Wang**, Elias Aboutanios. “Theoretical analysis of reconfigurable adaptive antenna array in GNSS applications.” *Proceedings of the 21th Eusipco*, Sep 2013, pp. 1-5.
3. **Xiangrong Wang**, Elias Aboutanios. “Adaptive beampattern synthesis with antenna selection by Iterative Shrinkage Continuation method.” *Proceedings of the 21th Eusipco*, Sep 2013, pp. 1-5.
4. **Xiangrong Wang**, Elias Aboutanios, Matthew Trinkle. “Subarray Selection for Adaptive Array Signal Processing in GNSS Applications.” *Proceedings of the 26th ION GNSS+*, Sep 2013, pp. 2776 - 2785.
5. **Xiangrong Wang**, Elias Aboutanios, Andrew Dempster. “Low Cost Adaptive Array Signal Processing by Subarray Selection.” *Proceedings of IGNSS Symposium*, Jul 2013.

6. **Xiangrong Wang**, Moeness Amin, Fauzia Ahmad, Elias Aboutanios. "Coarray Based Optimum Geometries for DOA Estimation with Multiple CRPA GPS Arrays." *Proceedings of the 27th ION GNSS+*, Sep 2014.
7. **Xiangrong Wang**, Elias Aboutanios, Moeness Amin. "Off-Grid High Resolution DOA Estimation for GNSS Circular Array Receivers." *Proceedings of the 27th ION GNSS+*, Sep 2014.
8. **Xiangrong Wang**, Moeness Amin, Fauzia Ahmad, Elias Aboutanios. "Minimum Output Power Anti-Jam with Fully Augmentable Arrays in GNSS." *Proceedings of the IEEE BenMAS*, Sep 2014.
9. **Xiangrong Wang**, Elias Aboutanios, Moeness Amin. "Generalised Array Reconfiguration for Adaptive Beamforming by Antenna Selection." *Proceedings of the 40th IEEE ICASSP*, Apr 2015.
10. **Xiangrong Wang**, Moeness Amin, Fauzia Ahmad, Elias Aboutanios. "Bayesian Compressive Sensing for DOA Estimation using the Difference Coarray." *Proceedings of the 40th IEEE ICASSP*, Apr 2015.
11. **Xiangrong Wang**, Moeness Amin, Xianbin Cao. "Optimum Adaptive Beamformer Design with Controlled Quiescent Pattern by Antenna Selection." *Proceedings of IEEE Radar Conference*, May 2017.
12. **Xiangrong Wang** and Moeness Amin. "Design of Optimum Sparse Array for Robust MVDR Beamforming Against Satellite DOA Mismatch." *Proceedings of IEEE CAMSAP Conference*, 2017.
13. **Xiangrong Wang**, Moeness Amin, Xianbin Cao. "Optimum Array Configurations of Maximum Output SNR for Quiescent Beamforming." *Proceedings of the 42th IEEE ICASSP*, Apr 2017.
14. **Xiangrong Wang**, Moeness Amin and Xianghua Wang. "Optimum Sparse Array Design for Multiple Beamformers with Common Receiver." *Proceedings of the 43th IEEE ICASSP*, Apr 2018.
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18. **Xiangrong Wang**, Moeness Amin and Elias Aboutanios. “Interferometric Array Design Under Regularized Antenna Placements for Interference Suppression.” *Proceedings of The 2018 IEEE 10th Sensor Array and Multichannel Signal Processing Workshop (SAM) (SAM2018)*, Jul 2018.
19. **Xiangrong Wang**, Huaiyuan Liang and Pengcheng Wang, “Detection and Tracking of UAVs using Interferometric Radar.” *Proceedings of IEEE International Radar Conference*, Sep 2019.
20. **Xiangrong Wang** and Qian Long, “Uncoupled FDA Beampattern Synthesis by Discrete Element Position and Frequency Offsets Pairing” *Proceedings of the 16th European Radar Conference*, Oct 2019.
21. Jing Xu and **Xiangrong Wang**, “Joint Radar-Communication System Design via FH Code Selection and PSK Modulation” *Proceedings of IEEE International Conference on Signal, Information and Data Processing (ICSIDP)*, Dec 2019.
22. Huaiyuan Liang, Pengcheng Wang and **Xiangrong Wang**, “Simultaneous Tracking of Multiple Targets using Interferometric FMCW Radar” *Proceedings of IEEE International Conference on Signal, Information and Data Processing (ICSIDP)*, Dec 2019.
23. Xuan Zhang and **Xiangrong Wang**, “Sparse Adaptive Beamformer Design with a Good Quiescent Beampattern” *Proceedings of IEEE International Conference on Signal, Information and Data Processing (ICSIDP)*, Dec 2019.
24. Yanan Ma, Xianbin Cao and **Xiangrong Wang**, “Efficient Off-grid DOA Estimation based on Modified MUSIC for Arbitrary Linear Arrays” *Proceedings of IEEE International Conference on Signal, Information and Data Processing (ICSIDP)*, Dec 2019.
25. **Xiangrong Wang**, and Elias Aboutanios. “Adaptive Reduced-Dimensional Beamspace Beamformer Design by Analogue Beam Selection.” *Proceedings of the 43th IEEE ICASSP*, May 2019.
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27. **Xiangrong Wang**, Pengcheng Wang, Victor C. Chen. “Simultaneous Measurement of Radial and Transversal Velocities Using a Dual-Frequency Interferometric Radar.” *Proceedings of IEEE Radar Conference*, Apr 2019.
28. Yanan Ma, Xianbin Cao, **Xiangrong Wang**. “Off-Grid DOA Estimation with Arbitrary-Shaped Linear Array Using Single Snapshot.” *Proceedings of IEEE Radar Conference*, Apr 2019.
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30. Huaiyuan Liang, Pengcheng Wang, **Xiangrong Wang**. “Transversal Velocity Measurement of Multiple Targets Based on Spatial Interferometric Averaging.” *Proceedings of International Radar Conference*, Apr 2020.
31. **Xiangrong Wang**, Maria Greco and Fulvio Gini. “Deterministic Sparse Array Capon Beamformer Design for Mitigating Spatially-Closed Interferences.” *Proceedings of IEEE Radar Conference*, Sep 2020.
32. Huaiyuan Liang, **Xiangrong Wang** and Alfonso Farina. “Performance Analysis and Optimal Placement for Unambiguous DOA Estimation using Interferometry.” *Proceedings of IEEE Radar Conference*, Sep 2020.
33. **Xiangrong Wang** and Aboulnasr Hassanien. “Phase Modulated Communications Embedded in Correlated FH-MIMO Radar Waveforms.” *Proceedings of IEEE Radar Conference*, Sep 2020.
34. Xin Zhang, **Xiangrong Wang** and Elias Aboutanios. “Effect Analysis of Spatial Modulation on Clutter Mitigation for Joint RadCom Systems and Solutions.” *Proceedings of IEEE Radar Conference*, Sep 2020.
35. Yanan Ma, Xianbin Cao and **Xiangrong Wang**. “Enhanced DOA Estimation for MIMO radar in the Case of Limited Snapshots.” *Proceedings of IEEE SAM Workshop*, Jun 2020.
36. **Xiangrong Wang**, Jing Xu, , Aboulnasr Hassanien and Elias Aboutanios. “Joint Communications with FH-MIMO Radar Systems: An Extended Signaling Strategy.” *Proceedings of 2021 IEEE International Conference on Acoustics, Speech and Signal*

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39. Weitong Zhai, **Xiangrong Wang**, Syed A. Hamza and Moeness G. Amin. “Cognitive Optimization of Sparse Array Transceiver for MIMO Radar Beamforming.” *Proceedings of IEEE Radar Conference*, May 2021.
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43. Weitong Zhai, **Xiangrong Wang**, Maria Greco and Fulvio Gini. “Weak Target Detection in Massive MIMO Radar via an Improved Reinforcement Learning Approach.” *Proceedings of 2022 IEEE International Conference on Acoustics, Speech and Signal Processing*, May 2022.
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46. Jing Xu, **Xiangrong Wang**, Xingshuai Qiao and Abdelhak M. Zoubir. "Sparse Array Design for Joint Communication Radar System via Antenna Selection." *Proceedings of International Symposium on Signals, Circuits & Systems - ISSCS 2023*, July 2023.
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51. Weitong Zhai, **Xiangrong Wang**, Moeness Amin, Maria S. Greco and Fulvio Gini. "IRS-Assisted Joint Sensing and Communication Design for Autonomous Driving." *Proceedings of 2024 ICASSP*, April 2024.
52. Kun Wu, **Xiangrong Wang**, Hengfeng Liu, Victor C. Chen and Elias Aboutanios, "Parameter Estimation of Rotary Drones in Far Distance using Long-Time Spectral Processing," 2023 IEEE International Radar Conference (RADAR), Sydney, Australia, 2023, pp. 1-6.
53. Weitong Zhai, Xiangrong Wang, Kunyan Li and Ziwei Wang, "Partially-Connected Hybrid Precoder Design for ISAC via Joint Grouping and Precoding", 2024 IEEE International Conference on Signal, Information and Data Processing, Zhuhai, China, 2024, **Best Paper Award**.
54. H. Liu, X. Wang, M. G. Amin and V. C. Chen, "Multi-Stroke Air-Writing Recognition Using Temporal-Spatial Interferometric MIMO Radar," 2024 IEEE Radar Conference (RadarConf24), Denver, CO, USA, 2024, doi: 10.1109/RadarConf2458775.2024.10548651.
55. H. Liu, **X. Wang**, M. Amin and Abdelhak M. Zoubir, "High-Quality Synthetic Data Generation For Omni-directional Human Activity Recognition," 2025 IEEE Interna-

tional Radar Conference, Atlanta, USA, 2025, **Best Student Paper Award Finalist**.

56. **X. Wang**, Maria S. Greco and Fulvio Gini, "Sparse Vector-Antenna Array Design for Dual-Functional Radar Communications," 2024 IEEE International Radar Conference, Rennes, France, 2024.
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